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The effect of minimum wage increases on inflation

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ABSTRACT

The wage level that employers cannot legally go below by setting a floor limit on wages is called the "minimum wage". In other words, the minimum wage is usually the lowest wage determined to meet the basic needs of workers. This wage is set to raise the living standards of workers and to reduce poverty. As a result of the increases in food prices in international markets in 2021 and especially in 2022, a food crisis emerged in the world economy in 2022. While the effects of the COVID-19 pandemic were overcome in 2021, the prices increased as a result of increased demand, on the other hand, at the beginning of 2022, the Russia-Ukraine tension triggered the food crisis while pushing food prices up. While rising prices affected fixed incomes the most, it became necessary to increase the minimum wage. Increasing the minimum wage encourages consumption by reducing poverty and increasing fixed income (workers) well-being. The increase in the minimum wage encourages consumption by reducing poverty and increases the welfare of the fixed income (workers), and on the one hand, increases the costs of the producers and leads to a decrease in production and unemployment. While increases can help improve workers' living standards, they can also have negative effects, such as reducing worker numbers or increasing costs for some businesses. The main purpose of this research is to examine that while increasing the minimum wage in the inflationary period is inevitable, it will again increase the general level of prices and trigger inflation.

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Introduction

In general, we can say that wage is the "reward of labor" or "price". Wages have different meanings for different economic units. For the worker, the wages of his labor and the income of himself and his family are the source of his livelihood. For employers, wage is a cost element and every wage increase is accepted as a cost increase.

For many years, the economic literature has been concerned with estimating the effects of minimum wage increases. In addition to being an economic policy, the minimum wage is also very important in its social dimension. The main purpose of the minimum wage, as a socio-economic policy aimed at fighting poverty and reducing income inequality, is to protect workers from unfairly low wages.

The minimum wage in North Macedonia was first introduced in 2012 with the adoption of the 'Minimum Wage Law'. In general, minimum wage adjustments have been made in March and April salaries every year. While there are many parameters in determining the minimum wage, purchasing power and announced inflation figures come first.

For a long time, high price increases were not seen in North Macedonia. As a matter of fact, the main goal of the Central Bank of North Macedonia since 1995 has been to ensure price stability. When we look at the inflation figures, we can see that the Central Bank has been successful in meeting this target.

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While the effects of the COVID 19 pandemic were overcome in 2021, the global food crisis that broke out in 2022 affected the whole world, while the North Macedonian economy experienced high price increases that have not been seen for many years. As a matter of fact, the country reached double-digit inflation in 2022.

While minimum wage increases provide an income increase in the fixed income group, it is useful to focus on the macroeconomic effects. In particular, the employment-unemployment relationship and its effect on inflation are important.

While many factors are effective in determining wages, living costs, in other words inflation, put pressure on wages. It is impossible for political governments to determine alone the wage level in the economy. Therefore, since the bargaining between the unions and the government take a long time, the wage levels are determined on an annual basis, especially the minimum wage. A high rate of inflation in an economy causes the real value of wages and salaries to fall. That is, less goods or services can be purchased with the same amount of money. The minimum wage determines the lowest wage level for workers and employees. In an economy with high inflation, the real value of the minimum wage decreases and the purchasing power of those working with minimum wage decreases. Therefore, it is important to make the minimum wage regulations taking into account the inflation rate. According to related studies, there is a direct and bidirectional relation between the inflation and the minimum wage.

The main purpose of this research is to examine that while it is inevitable to increase the minimum wage in the inflationary period, it will again increase the general level of prices and trigger inflation.

Literature Review

Although there are many studies on the economic effects of the minimum wage, researches on the relationship between the minimum wage and inflation remain limited. There are hardly any such studies in North Macedonia. In general, it can be understood that the minimum wage has a direct relationship with inflation.

In the OECD, Minimum wages in times of rising inflation (2022) research, while emphasizing the importance of carefully addressing both the economic and social effects of minimum wages, it was expressed to be careful about raising the minimum wages. In this direction, it is important to turn to special and temporary solutions in order to avoid the wage-price spiral.

Majchrowska, A. (2022) we analyzed the pass-through effect of the minimum wage on inflation in 16 regional labor markets in Poland in 2003–2020. The results show that this effect is statistically significant and positive. Increases in the minimum-average wage rate in the analyzed period contributed positively to the rise in inflation. Also, the minimum wage pass-through effect was higher in this case. This is in line with economic theory and confirms previous empirical findings. Sevinç, D. (2022), when the results of the analysis are evaluated, it can be said that the results show slight differences due to the taxes and deductions in the cost of the gross minimum wage and the minimum wage to the employer. The demand for goods and services resulting from the increase rate at the net minimum wage causes an increase the food inflation, which is one of the biggest consumption items of minimum wage earners, not in general consumer inflation. Ashenfelter, Orley C. and Jurajda, Stepan, Wages (2021) the research analyzed the ability of McDonald's restaurants to reflect higher labor outcomes caused by minimum wage results in the form of higher Big Mac sandwich prices between 2016 and 2020. Pay levels at McDonald's restaurants, the largest franchise chain in the US, are strongly influenced by minimum wage legislation. In addition, almost all such labor costs for organizations are reflected in higher Big Mac prices.

In Akgül, O. & Bükey, A. M. (2020) research study, a long-term relationship between the annual minimum wage and inflation in Turkey's 1987-2018 reference range was determined by means of ARDL bounds test approach and a bidirectional relationship was determined by Toda-Yamamoto Causality analysis. Brummund and Strain (2020) suggest that since changing the composition of production factors is costly, firms may decide not to respond to sporadic (non-permanent) increases in the nominal minimum wage that are expected to be completely eroded by inflation in the long run.

In the study of Tüleykan, H. (2019), it was concluded that minimum wage increases will create inflationary pressure in the long run. It is recommended that all aspects of the policies be implemented through intervention in the minimum wage be thoroughly evaluated. Knotek & Zaman (2014) analysis reveals that wages and prices tend to move together, complicating efforts to resolve cause and effect. The slow rise of wages reveals the links between wages, prices and economic activity. Hoxha, A. (2010). Despite most of the studies usually giving one way causality this study of the price-wage causal relationship in EU-12 supports the case of bilateral causality.

The Impact of the Minimum Wage Increase on The Market

Before moving on to the Figure 1 analysis below, it is worth emphasizing the assumption that all other costs in the economy remain constant during the analysis.

Initially, the balance in the economy was achieved at the point (E) where supply and demand were equalized. At this point, while the market price is 40 units, the quantity of goods produced and sold is 35 units. An increase in the minimum wage will increase market demand and the demand curve will shift from D to D1. At this point, the price will increase and due to the increase in demand, the new balance will be achieved at the level E1 = Supply (S) = Demand (D1) = P (50) = Q (40). On the other hand, the 12.56 percent increase in the minimum wage in 2023 will trigger an increase in the wages of other wage earners, which will increase the labor cost of the companies. Companies with increasing costs will reduce the number of employees, and as a result, the amount of production

will be decreased. As a result of the decrease in production, the market supply curve will shift from S to S1, which will move prices upwards. As a result of the decrease in supply, the balance level will change again and the new balance will be achieved at the level E2 = Supply (S1) = Demand (D1) = P (60) = Q (35) (Dervis, 2021).

As a result, since the increase in demand will increase the price, while the decrease in supply will increase the prices, the balanc price level will increase twice due to both the increase in demand and decrease in supply, the amount of production will increase at first, as a result of the decrease in supply, will be decreased and returned at the initial level.

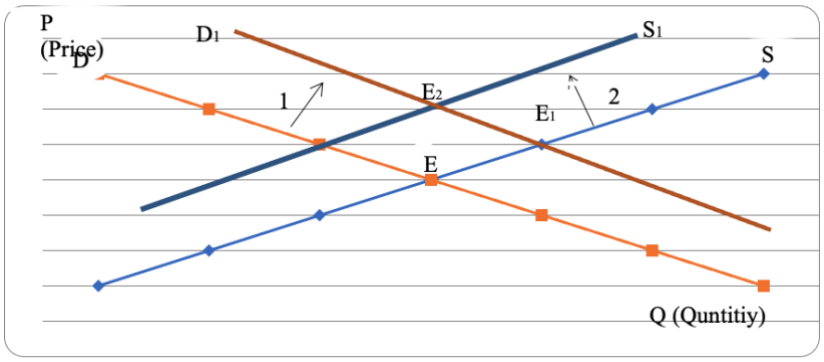

Figure 1: The Impact of The Minimum Wage Increase on The Market

The Position of The Minimum Wage Against Inflation in North Macedonia

The Figure below contain the minimum wage data for the years 2020-2023 and the inflation data for the years 2019-2022. The minimum wage in North Macedonia with the adoption of the 'Minimum Wage Law', was first introduced in 2012. In the last 12 years, the minimum wage in the country has increased by 137 percent.

Table 1: The Minimum Wage Data of North Macedonia

Year	Net (Denar)	Gross (Denar)	Variation (%)
2020	14934	21776	
2021	15194	22146	1,70
2022	18000	26422	19,31
2023	20175	29740	12,56

Source: North Macedonian Statistical Institute, Central Bank of North Macedonian

The main objective of the Central Bank of North Macedonia set by law, is to ensure price stability. To achieve the final target, the central bank sets the intermediate target of monetary policy. Thus, from April 1992 to September 1995, the central bank implemented the strategy of targeting the M1 money supply as an "intermediate target" of monetary policy. Since October 1995, the central bank has been implementing a fixed exchange rate strategy of the Macedonian denar against the German mark, and from January 2002 against the Euro. Therefore, ensuring the stability of the exchange rate is one of the intermediate objectives of monetary policy.

Table 2: Inflation Data of North Macedonia

	Consumer price index					
Year	Total	Food and soft drinks	Housing, water, electricity, gas and other fuels	Transport	**Retail prices**	**Industry and producer prices**
2019	0,8	1,6	-0,1	-2,2	0,0	0,9
2020	1,2	2,4	2,0	-5,9	0,4	0,8
2021	3,2	3,1	2,8	10,0	4,2	8,5
2022	14,2	20,9	13,3	19,5	12,6	21,9

Source: North Macedonian Statistical Institute, Central Bank of North Macedonian

There are basically two sources of the increase in the general level of prices. The first is demand inflation, which occurs due to demand, and the other is cost inflation, which is caused by supply. The price increases in the country in recent years are due to the increase in costs. While increasing production costs increase prices, it also decreases purchasing power. While the purchasing power

of money depreciates against inflation, it is necessary to compensate for this with wage increases. As a matter of fact, while the inflation rate was below 5 percent in 2020 and 2021, the minimum wage increase remained at the same level in the same period.

The demand and supply gap(imbalance) that occurred during the COVID 19 pandemic process, the raw material prices have increased due to transportation problems and problems in production. In addition, the effect of the upward movements in energy and fuel prices was reflected on the final products. While all these developments were taking place, the Russia-Ukraine tensions at the beginning of 2022 pushed the food prices upwards and the inflation rates increased throughout the world.

While these developments are taking place in the world, the economy of North Macedonia has experienced price increases in 2022, which it has not seen in the last 25 years. While the inflation rate was 3.2 in 2021, it increased to 14.2 in 2022. In the same year, food and non-alcoholic beverage prices increased by 20.9 percent, and industrial and producer prices increased by 21.9 percent.

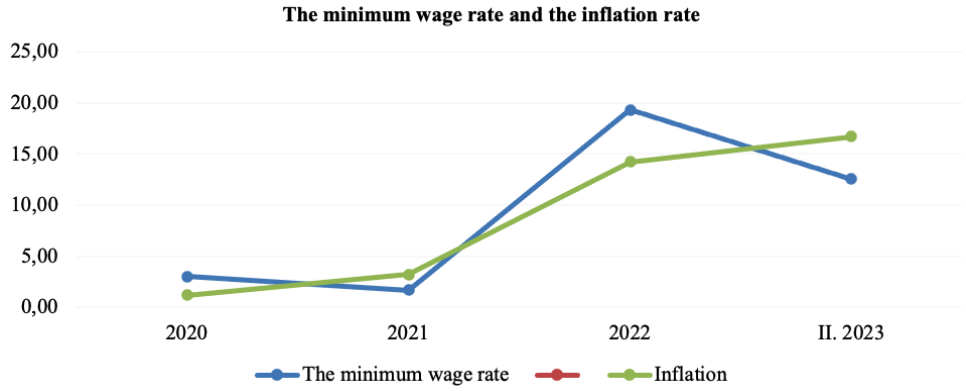

Figure 2: The Course of The Minimum Wage Rate and The Inflation Rate; *Source:* Processed Primary Data 2023

When we look at the situation of the minimum wage against inflation, the percent increase in the minimum wage in 2020 and 2021 is compatible with inflation, while the percentage increase in the minimum wage in 2020 and 2021 is consistent with inflation, as the inflation rate was 14.2 percent in 2022, the minimum wage increased by approximately 20 percent in the same year. As of March 2023, the minimum wage in the country has increased by 12.56 percent. As of February 2023, the inflation rate was realized as 16.7 percent. We can see that the annual increase in the minimum wage remains below the inflation rate.

Conclusions

In case of an increase in inflation, there is a decrease in the real purchasing power of those with fixed incomes. This results in those with the same income receiving fewer goods and services. Therefore, it is inevitable for governments to increase the minimum wage in inflationary periods. In inflationary periods, the minimum wage increase has both positive and negative effects.

In the above study, the theoretical effect of the recent inflation and minimum wage increases in North Macedonia on the market and its current situation were analyzed.

According to literature studies, there is a bidirectional relationship between increasing the minimum wage and inflation, while OECD studies reveal the need to be careful about raising minimum wages.

Single-digit inflation was seen in North Macedonia from 1995 to 2021. With the end of the effects of the COVID 19 pandemic in 2022, the demand for goods and services in the economy has increased. On the other hand, at the beginning of 2022, the Russia-Ukraine crisis especially affected the food sector and a contraction in the supply side of the economy was experienced. The increase in demand on the one hand and the decrease in supply on the other hand increased the general level of prices in the country in 2022. After a long period, North Macedonia faced the double-digit inflation rate.

These developments in the world economy have directly affected the economy of the country. In the face of increasing inflation, the North Macedonian government increased the minimum wages by 19.3 percent in 2022 and by 12.5 percent in 2023. The increase in the minimum wage above the inflation rate in 2022 led to an increase in other wages. Wage increases have increased the purchasing power of fixed incomes and triggered the increase in total demand in the economy. This contributed to the increase in the general level of prices and the increase in the inflation rate. When we look at the supply side of the economy, an increase in the minimum wage will increase the production costs of the producers, causing an increase in the prices of goods and services on the one hand, and a decrease in production on the other, which will increase inflation. As a result, an increase in the minimum wage will increase the total demand in the economy and increase inflation, and as a result of the increase in production costs, prices and inflation will increase again.

It is difficult to make the decision to increase the minimum wage during the inflationary period and many factors should be taken into account while making the decision. However, it is necessary in the inflationary period, to maintain economic stability by using the minimum wage increase in a balanced manner with other policy tools.

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Institutional Review Board Statement: Ethical review and approval were waived for this study, due to that the research does not deal with vulnerable groups or sensitive issues.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy.

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